Cardiotoxicity and Chemotherapy or Radiation Treatment

Quick Tips – Care Guide

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From initial sarcoma diagnosis to survivorship, a multidisciplinary approach involving cardio-oncology providers should be adopted to minimize cardiotoxicity and ensure long-term cardiovascular health.

- **Sarcoma survivors exposed to anthracyclines** should be referred to a comprehensive survivorship clinic that includes **cardio-oncology evaluation**.
- For patients deemed to be at **moderate or high risk of cardiac dysfunction**, intensive management of cardiovascular risk factors is essential, including control of **blood pressure, lipids, and diabetes**.
- Even in sarcomas that do not directly involve the heart, **cardiotoxicity** must be considered throughout treatment and into survivorship because of the near-universal use of **high-dose anthracyclines**.
- Other cytotoxic therapies contribute less to cardiac dysfunction, but **TKIs** and **immunotherapies** have unique cardiovascular side-effect profiles that must also be considered.
- In particular, the onset of **hypertension** (with TKIs) and **myocarditis** (with ICIs) should be closely monitored in clinical practice to allow for early intervention.
- The sarcoma community must remain vigilant—especially for **pediatric patients** and those predisposed to cardiac disease because of comorbidities—and **long-term monitoring is recommended**.

Cancer treatments with **cardiotoxic potential** can increase the risk of **heart failure, irregular heart rhythms, and heart attacks**. In addition, radiation to the chest for **breast, lung, and other cancers** can damage heart tissue, including the **heart valves and arteries**.

If you are a **cancer or sarcoma patient** at risk for treatment-related heart problems, your oncologist may work with a cardiologist to evaluate your risk and help reduce potential cardiotoxic effects, says **Zeeshan Hussain, MD**, a cardiologist at **UH Harrington Heart & Vascular Institute**.

Here, Dr. Hussain answers common questions he hears from patients about cardiotoxicity:

Will my oncologist refer me to a cardiologist if needed, or should I ask to see a cardiologist before starting therapy?

Patients with existing heart disease, those at increased risk for heart disease and those who will receive potentially cardiotoxic therapies should be evaluated before their treatment and followed by a cardiologist. Most oncologists are aware of the need for referrals.

What questions should I be asking?

You should ask if you're at increased cardiac risk and if the treatment is cardiotoxic. Also ask what can be done to reduce cardiac risk from this treatment.

What are symptoms of cardiotoxicity? What steps can I take to reduce my risk?

Shortness of breath, chest pain, palpitations, lightheadedness could all be signs or symptoms of cardiotoxicity. Particularly with cancer drugs, cardiotoxicity can happen without overt symptoms. Therefore, it is important to have periodic assessments if you are on such medications.

There are certain medications that can reduce future risk of cardiotoxicity. Maintaining a healthy weight and keeping your blood pressure and cholesterol under control are also good preventative strategies.

What type of heart function monitoring should I receive – and how often?

The heart can be monitored in a variety of ways before, during and after cancer treatment, if you are receiving radiation or chemotherapy that is potentially cardiotoxic. We use ultrasound with innovative and sensitive methods to quantify heart function. Biomarkers (i.e., blood tests) are also used for surveillance. If additional information regarding the amount of scar or abnormal proteins in the heart is required, a cardiac magnetic resonance imaging may also be considered. The exact type of test, and the frequency with which someone needs to be monitored, is typically at the discretion of the treating cardiologist.

Should have my heart function checked even though it’s been years since I underwent cancer treatment?

Certain chemotherapeutic agents carry a risk of cardiotoxicity that extend way beyond the duration of treatment. Particularly with radiation to the heart, the changes can manifest many years later and necessitate long-term follow-up.

Resources

National LMS Foundation

Cleveland Clinic

Cardiovascular Risk Report

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.org

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